

API Request and Response Formats

Quick Reference

Total APIs: 69 Endpoints

Module	Endpoints	Description
Authentication	3	Login, logout, token refresh
Users	8	Registration, profile management, user administration
Facilities	7	Facility CRUD, types, availability
Equipment	7	Equipment CRUD, categories, availability
Reservations	7	Reservation management, approval workflow
Borrowing	8	Borrowing management, approval, return workflow
Notifications	3	User notifications, read status
Reports & Logs	3	Dashboard, audit logs, overdue items
Lookups	6	Reference data (statuses, roles, types)
Additional Features	5	Calendar, search, extensions, activity logs

User Roles & Permissions

Admin (role_id: 1)

- Full system access
- Manage facilities and equipment
- Approve/decline reservations and borrowing requests
- View all users, logs, and reports
- Generate reports and analytics

Organization (role_id: 2), Student (role_id: 3)

- Create and manage reservations
- Create and manage borrowing requests
- View own request history
- Receive notifications

API Organization

Authentication

- POST /api/auth/login
- POST /api/auth/logout
- POST /api/auth/refresh-token`

Users

- POST /api/users/register/student
- POST /api/users/register/organization
- POST /api/users/register/admin
- GET /api/users/profile
- PUT /api/users/profile
- DELETE /api/users/:id
- GET /api/users (Admin only)
- GET /api/users/:id (Admin only)
- PATCH /api/users/:id/status (Admin only)

Facilities

- GET /api/facilities
- GET /api/facilities/:id
- POST /api/facilities (Admin only)
- PUT /api/facilities/:id (Admin only)
- DELETE /api/facilities/:id (Admin only)
- GET /api/facilities/:id/availability
- GET /api/facility-types

Equipment

- GET /api/equipment
- GET /api/equipment/:id
- POST /api/equipment (Admin only)
- PUT /api/equipment/:id (Admin only)
- DELETE /api/equipment/:id (Admin only)
- GET /api/equipment/:id/availability
- GET /api/equipment-categories

Reservations

- GET /api/reservations
- GET /api/reservations/:id

- POST /api/reservations
- PATCH /api/reservations/:id
- DELETE /api/reservations/:id
- PATCH /api/reservations/:id/approve (Admin only)
- PATCH /api/reservations/:id/decline (Admin only)
- GET /api/reservations/history

Borrowing

- GET /api/borrowing
- GET /api/borrowing/:id
- POST /api/borrowing
- PATCH /api/borrowing/:id
- DELETE /api/borrowing/:id
- PATCH /api/borrowing/:id/approve (Admin only)
- PATCH /api/borrowing/:id/decline (Admin only)
- PATCH /api/borrowing/:id/return (Admin only)
- GET /api/borrowing/history

Notifications

- GET /api/notifications
- PATCH /api/notifications/:id/read
- PATCH /api/notifications/read-all

Reports & Logs

- GET /api/reports/dashboard
- GET /api/reports/audit-logs
- GET /api/reports/overdue

Lookups

- GET /api/lookups/roles
- GET /api/lookups/user-statuses
- GET /api/lookups/reservation-statuses
- GET /api/lookups/borrowing-statuses

AUTHENTICATION

POST /api/auth/login

Request:

```
json
{
  "email": "string (required)",
  "password": "string (required)"
}
```

Response (200 OK):

```
json
{
  "success": true,
  "message": "Login successful",
  "data": {
    "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
    "user": {
      "user_id": 1,
      "email": "john.doe@example.com",
      "first_name": "John",
      "last_name": "Doe",
      "phone_number": "+639123456789",
      "role_id": 3,
      "role_name": "Student",
      "status_id": 1,
      "status_name": "Active",
      "created_at": "2025-01-15T08:30:00Z"
    }
  }
}
```

Response (401 Unauthorized):

```
json
{
  "success": false,
  "message": "Invalid email or password",
}
```

```
"data": null
}
```

POST /api/auth/logout

Request:

```
json
{
  "token": "string (required)"
}
```

Response (200 OK):

```
json
{
  "success": true,
  "message": "Logout successful",
  "data": null
}
```

POST /api/auth/refresh-token

Request:

```
json
{
  "refresh_token": "string (required)"
}
```

Response (200 OK):

```
json
{
  "success": true,
  "message": "Token refreshed successfully",
  "data": {
    "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
    "refresh_token": "new_refresh_token_here"
  }
}
```

```
}
```

USERS

POST /api/users/register/student

Request:

```
json
{
  "email": "string (required)",
  "password": "string (required, min 8 characters)",
  "first_name": "string (required)",
  "last_name": "string (required)",
  "phone_number": "string (optional)"
}
```

Response (201 Created):

```
json
{
  "success": true,
  "message": "Student registered successfully",
  "data": {
    "user_id": 15,
    "email": "student@example.com",
    "first_name": "Jane",
    "last_name": "Smith",
    "phone_number": "+639123456789",
    "role_id": 3,
    "role_name": "Student",
    "status_id": 1,
    "status_name": "Active",
    "created_at": "2025-10-09T10:30:00Z"
  }
}
```

Response (400 Bad Request):

```
json
{
  "success": false,
  "message": "Email already exists",
}
```

```
"data": null
}
```

POST /api/users/register/organization

Request:

```
json
{
  "email": "string (required)",
  "password": "string (required, min 8 characters)",
  "organization_name": "string (required)",
  "phone_number": "string (optional)"
}
```

Response (201 Created):

```
json
{
  "success": true,
  "message": "Organization registered successfully",
  "data": {
    "user_id": 16,
    "email": "org@example.com",
    "first_name": "Tech Club",
    "last_name": null,
    "phone_number": "+639123456789",
    "role_id": 2,
    "role_name": "Organization",
    "status_id": 1,
    "status_name": "Active",
    "created_at": "2025-10-09T10:30:00Z"
  }
}
```

POST /api/users/register/admin

Request:

```
json
```

```
{
  "email": "string (required)",
  "password": "string (required, min 8 characters)",
  "first_name": "string (required)",
  "last_name": "string (required)",
  "phone_number": "string (optional)"
}
```

Response (201 Created):

```
json
{
  "success": true,
  "message": "Admin registered successfully",
  "data": {
    "user_id": 17,
    "email": "admin@example.com",
    "first_name": "Admin",
    "last_name": "User",
    "phone_number": "+639123456789",
    "role_id": 1,
    "role_name": "Admin",
    "status_id": 1,
    "status_name": "Active",
    "created_at": "2025-10-09T10:30:00Z"
  }
}
```

GET /api/users/profile

Request:

Headers:

Authorization: Bearer {token}

Response (200 OK):

```
json
{
  "success": true,
  "message": "Profile retrieved successfully",
  "data": {
```

```
"user_id": 15,  
"email": "student@example.com",  
"first_name": "Jane",  
"last_name": "Smith",  
"phone_number": "+639123456789",  
"role_id": 3,  
"role_name": "Student",  
"status_id": 1,  
"status_name": "Active",  
"created_at": "2025-10-09T10:30:00Z",  
"updated_at": "2025-10-09T14:20:00Z"  
}  
}
```

PUT /api/users/profile

Request:

```
json  
{  
  "first_name": "string (optional)",  
  "last_name": "string (optional)",  
  "phone_number": "string (optional)",  
  "current_password": "string (required if changing password)",  
  "new_password": "string (optional)"  
}
```

Response (200 OK):

```
json  
{  
  "success": true,  
  "message": "Profile updated successfully",  
  "data": {  
    "user_id": 15,  
    "email": "student@example.com",  
    "first_name": "Jane",  
    "last_name": "Doe",  
    "phone_number": "+639987654321",  
    "role_id": 3,  
    "role_name": "Student",  
    "status_id": 1,  

```



```
"status_name": "Active",  
"updated_at": "2025-10-09T15:45:00Z"  
}  
}
```

DELETE /api/users/:id

Request:

Path Parameter:

id: integer (required)

Headers:

Authorization: Bearer {token}

Response (200 OK):

```
json  
{  
  "success": true,  
  "message": "User deleted successfully",  
  "data": null  
}
```

GET /api/users (*Admin only*)

Request:

Query Parameters:

page: integer (optional, default: 1)
limit: integer (optional, default: 10)
role_id: integer (optional)
status_id: integer (optional)
search: string (optional, searches email/name)

Headers:

Authorization: Bearer {admin_token}

Response (200 OK):

```
json
{
  "success": true,
  "message": "Users retrieved successfully",
  "data": {
    "users": [
      {
        "user_id": 15,
        "email": "student@example.com",
        "first_name": "Jane",
        "last_name": "Smith",
        "phone_number": "+639123456789",
        "role_id": 3,
        "role_name": "Student",
        "status_id": 1,
        "status_name": "Active",
        "created_at": "2025-10-09T10:30:00Z"
      },
      {
        "user_id": 16,
        "email": "org@example.com",
        "first_name": "Tech Club",
        "last_name": null,
        "phone_number": "+639123456780",
        "role_id": 2,
        "role_name": "Organization",
        "status_id": 1,
        "status_name": "Active",
        "created_at": "2025-10-08T09:15:00Z"
      }
    ],
    "pagination": {
      "current_page": 1,
      "total_pages": 5,
      "total_users": 48,
      "per_page": 10
    }
  }
}
```

GET /api/users/:id (Admin only)

Request:

Path Parameter:

id: integer (required)

Headers:

Authorization: Bearer {admin_token}

Response (200 OK):

```
json
{
  "success": true,
  "message": "User retrieved successfully",
  "data": {
    "user_id": 15,
    "email": "student@example.com",
    "first_name": "Jane",
    "last_name": "Smith",
    "phone_number": "+639123456789",
    "role_id": 3,
    "role_name": "Student",
    "status_id": 1,
    "status_name": "Active",
    "created_at": "2025-10-09T10:30:00Z",
    "updated_at": "2025-10-09T14:20:00Z",
    "total_reservations": 12,
    "total_borrowings": 5,
    "active_reservations": 2,
    "active_borrowings": 1
  }
}
```

PATCH /api/users/:id/status (Admin only)**Request:**

```
json
{
  "status_id": "integer (required, 1=Active, 2=Inactive, 3=Suspended)"
}
```

Response (200 OK):

```
json
{
  "success": true,
  "message": "User status updated successfully",
  "data": {
    "user_id": 15,
    "status_id": 2,
    "status_name": "Inactive",
    "updated_at": "2025-10-09T16:00:00Z"
  }
}
```

FACILITIES

GET /api/facilities

Request:

Query Parameters:

page: integer (optional, default: 1)
limit: integer (optional, default: 10)
facility_type_id: integer (optional)
status_id: integer (optional)
search: string (optional)

Response (200 OK):

```
json
{
  "success": true,
  "message": "Facilities retrieved successfully",
  "data": {
    "facilities": [
      {
        "facility_id": 1,
        "facility_name": "Conference Room A",
        "description": "Large conference room with projector",
        "facility_type_id": 1,
        "facility_type_name": "Conference Room",
        "capacity": 50,
        "status_id": 1,

```

```
    "status_name": "Available",
    "created_at": "2025-01-10T08:00:00Z",
    "updated_at": "2025-10-01T10:00:00Z"
  },
  {
    "facility_id": 2,
    "facility_name": "Basketball Court",
    "description": "Outdoor basketball court",
    "facility_type_id": 3,
    "facility_type_name": "Sports Facility",
    "capacity": 20,
    "status_id": 1,
    "status_name": "Available",
    "created_at": "2025-01-10T08:00:00Z",
    "updated_at": "2025-10-01T10:00:00Z"
  }
],
"pagination": {
  "current_page": 1,
  "total_pages": 3,
  "total_facilities": 25,
  "per_page": 10
}
}
```

GET /api/facilities/:id

Request:

Path Parameter:

id: integer (required)

Response (200 OK):

```
json
{
  "success": true,
  "message": "Facility retrieved successfully",
  "data": {
    "facility_id": 1,
    "facility_name": "Conference Room A",
```

```
"description": "Large conference room with projector and whiteboard",
"facility_type_id": 1,
"facility_type_name": "Conference Room",
"capacity": 50,
"status_id": 1,
"status_name": "Available",
"created_at": "2025-01-10T08:00:00Z",
"updated_at": "2025-10-01T10:00:00Z",
"upcoming_reservations": [
  {
    "reservation_id": 45,
    "reservation_date": "2025-10-10",
    "start_time": "09:00:00",
    "end_time": "11:00:00",
    "user_name": "Jane Smith"
  }
]
}
```

POST /api/facilities (Admin only)

Request:

```
json
{
  "facility_name": "string (required)",
  "description": "string (optional)",
  "facility_type_id": "integer (required)",
  "capacity": "integer (optional)",
  "status_id": "integer (optional, default: 1)"
}
```

Response (201 Created):

```
json
{
  "success": true,
  "message": "Facility created successfully",
  "data": {
    "facility_id": 26,
    "facility_name": "Study Room B",

```

```
"description": "Small study room for groups",
"facility_type_id": 2,
"facility_type_name": "Study Room",
"capacity": 10,
"status_id": 1,
"status_name": "Available",
"created_at": "2025-10-09T16:30:00Z"
}
}
```

PUT /api/facilities/:id (Admin only)

Request:

```
json
{
  "facility_name": "string (optional)",
  "description": "string (optional)",
  "facility_type_id": "integer (optional)",
  "capacity": "integer (optional)",
  "status_id": "integer (optional)"
}
```

Response (200 OK):

```
json
{
  "success": true,
  "message": "Facility updated successfully",
  "data": {
    "facility_id": 1,
    "facility_name": "Conference Room A - Updated",
    "description": "Large conference room with new equipment",
    "facility_type_id": 1,
    "facility_type_name": "Conference Room",
    "capacity": 60,
    "status_id": 1,
    "status_name": "Available",
    "updated_at": "2025-10-09T17:00:00Z"
  }
}
```

DELETE /api/facilities/:id (Admin only)

Request:

Path Parameter:

id: integer (required)

Headers:

Authorization: Bearer {admin_token}

Response (200 OK):

```
json
{
  "success": true,
  "message": "Facility deleted successfully",
  "data": null
}
```

Response (400 Bad Request - Has active reservations):

```
json
{
  "success": false,
  "message": "Cannot delete facility with active reservations",
  "data": {
    "active_reservations": 3
  }
}
```

GET /api/facilities/:id/availability

Request:

Path Parameter:

id: integer (required)

Query Parameters:

date: string (required, format: YYYY-MM-DD)

start_time: string (optional, format: HH:MM:SS)

end_time: string (optional, format: HH:MM:SS)

Response (200 OK):

```
json
{
  "success": true,
  "message": "Facility availability retrieved successfully",
  "data": {
    "facility_id": 1,
    "facility_name": "Conference Room A",
    "date": "2025-10-10",
    "is_available": false,
    "occupied_slots": [
      {
        "reservation_id": 45,
        "start_time": "09:00:00",
        "end_time": "11:00:00",
        "reserved_by": "Jane Smith"
      },
      {
        "reservation_id": 48,
        "start_time": "14:00:00",
        "end_time": "16:00:00",
        "reserved_by": "Tech Club"
      }
    ],
    "available_slots": [
      {
        "start_time": "08:00:00",
        "end_time": "09:00:00"
      },
      {
        "start_time": "11:00:00",
        "end_time": "14:00:00"
      },
      {
        "start_time": "16:00:00",
        "end_time": "18:00:00"
      }
    ]
  }
}
```

GET /api/facility-types

Request:

No parameters required

Response (200 OK):

```
json
{
  "success": true,
  "message": "Facility types retrieved successfully",
  "data": [
    {
      "facility_type_id": 1,
      "type_name": "Conference Room",
      "description": "Rooms for meetings and conferences"
    },
    {
      "facility_type_id": 2,
      "type_name": "Study Room",
      "description": "Small rooms for individual or group study"
    },
    {
      "facility_type_id": 3,
      "type_name": "Sports Facility",
      "description": "Outdoor and indoor sports facilities"
    },
    {
      "facility_type_id": 4,
      "type_name": "Laboratory",
      "description": "Computer and science laboratories"
    },
    {
      "facility_type_id": 5,
      "type_name": "Auditorium",
      "description": "Large halls for events and presentations"
    }
  ]
}
```

EQUIPMENT

GET /api/equipment

Request:

Query Parameters:

page: integer (optional, default: 1)
limit: integer (optional, default: 10)
category_id: integer (optional)
status_id: integer (optional)
search: string (optional)

available_only: boolean (optional, default: false)

Response (200 OK):

```
json
{
  "success": true,
  "message": "Equipment retrieved successfully",
  "data": {
    "equipment": [
      {
        "equipment_id": 1,
        "equipment_name": "Projector",
        "equipment_code": "PROJ-001",
        "category_id": 1,
        "category_name": "Audio Visual",
        "brand_model": "Epson EB-X05",
        "total_quantity": 10,
        "available_quantity": 7,
        "status_id": 1,
        "status_name": "Available",
        "specification": "3300 lumens, XGA resolution",
        "created_at": "2025-01-10T08:00:00Z",
        "updated_at": "2025-10-09T10:00:00Z"
      },
      {
        "equipment_id": 2,
        "equipment_name": "Laptop",
        "equipment_code": "LAP-001",
        "category_id": 2,
        "category_name": "Computing",
```

```
    "brand_model": "Dell Latitude 5420",
    "total_quantity": 15,
    "available_quantity": 12,
    "status_id": 1,
    "status_name": "Available",
    "specification": "Intel i5, 8GB RAM, 256GB SSD",
    "created_at": "2025-01-10T08:00:00Z",
    "updated_at": "2025-10-09T10:00:00Z"
  }
],
"pagination": {
  "current_page": 1,
  "total_pages": 4,
  "total_equipment": 38,
  "per_page": 10
}
}
```

GET /api/equipment/:id

Request:

Path Parameter:

id: integer (required)

Response (200 OK):

```
json
{
  "success": true,
  "message": "Equipment retrieved successfully",
  "data": {
    "equipment_id": 1,
    "equipment_name": "Projector",
    "equipment_code": "PROJ-001",
    "category_id": 1,
    "category_name": "Audio Visual",
    "brand_model": "Epson EB-X05",
    "total_quantity": 10,
    "available_quantity": 7,
    "status_id": 1,
```

```
"status_name": "Available",
"specification": "3300 lumens, XGA resolution, HDMI and VGA ports",
"created_at": "2025-01-10T08:00:00Z",
"updated_at": "2025-10-09T10:00:00Z",
"current_borrowings": [
  {
    "borrowing_id": 23,
    "user_name": "Jane Smith",
    "borrow_date": "2025-10-08",
    "expected_return_date": "2025-10-12",
    "status_name": "Approved"
  }
]
}
```

POST /api/equipment (*Admin only*)

Request:

```
json
{
  "equipment_name": "string (required)",
  "equipment_code": "string (optional)",
  "category_id": "integer (required)",
  "brand_model": "string (optional)",
  "total_quantity": "integer (required)",
  "available_quantity": "integer (required)",
  "status_id": "integer (optional, default: 1)",
  "specification": "text (optional)"
}
```

Response (201 Created):

```
json
{
  "success": true,
  "message": "Equipment created successfully",
  "data": {
    "equipment_id": 39,
    "equipment_name": "Microphone",
    "equipment_code": "MIC-001",

```

```
"category_id": 1,
"category_name": "Audio Visual",
"brand_model": "Shure SM58",
"total_quantity": 5,
"available_quantity": 5,
"status_id": 1,
"status_name": "Available",
"specification": "Dynamic vocal microphone",
"created_at": "2025-10-09T17:30:00Z"
}
}
```

PUT /api/equipment/:id (Admin only)

Request:

```
json
{
  "equipment_name": "string (optional)",
  "equipment_code": "string (optional)",
  "category_id": "integer (optional)",
  "brand_model": "string (optional)",
  "total_quantity": "integer (optional)",
  "available_quantity": "integer (optional)",
  "status_id": "integer (optional)",
  "specification": "text (optional)"
}
```

Response (200 OK):

```
json
{
  "success": true,
  "message": "Equipment updated successfully",
  "data": {
    "equipment_id": 1,
    "equipment_name": "Projector - HD",
    "equipment_code": "PROJ-001",
    "category_id": 1,
    "category_name": "Audio Visual",
    "brand_model": "Epson EB-X05",
    "total_quantity": 12,

```

```
"available_quantity": 9,  
"status_id": 1,  
"status_name": "Available",  
"specification": "3300 lumens, Full HD resolution",  
"updated_at": "2025-10-09T18:00:00Z"  
}  
}
```

DELETE /api/equipment/:id (Admin only)

Request:

Path Parameter:

id: integer (required)

Headers:

Authorization: Bearer {admin_token}

Response (200 OK):

```
json  
{  
  "success": true,  
  "message": "Equipment deleted successfully",  
  "data": null  
}
```

Response (400 Bad Request - Has active borrowings):

```
json  
{  
  "success": false,  
  "message": "Cannot delete equipment with active borrowings",  
  "data": {  
    "active_borrowings": 2  
  }  
}
```

GET /api/equipment/:id/availability

Request:

Path Parameter:

id: integer (required)

Query Parameters:

start_date: string (required, format: YYYY-MM-DD)

end_date: string (required, format: YYYY-MM-DD)

Response (200 OK):

json

```
{
  "success": true,
  "message": "Equipment availability retrieved successfully",
  "data": {
    "equipment_id": 1,
    "equipment_name": "Projector",
    "total_quantity": 10,
    "available_quantity": 7,
    "date_range": {
      "start_date": "2025-10-10",
      "end_date": "2025-10-15"
    },
    "daily_availability": [
      {
        "date": "2025-10-10",
        "borrowed_quantity": 3,
        "available_quantity": 7
      },
      {
        "date": "2025-10-11",
        "borrowed_quantity": 5,
        "available_quantity": 5
      },
      {
        "date": "2025-10-12",
        "borrowed_quantity": 2,
        "available_quantity": 8
      }
    ],
    "current_borrowings": [
      {
        "borrowing_id": 23,
```



```
    "user_name": "Jane Smith",
    "borrow_date": "2025-10-08",
    "expected_return_date": "2025-10-12"
  }
]
}
}
```

GET /api/equipment-categories

Request:

No parameters required

Response (200 OK):

```
json
{
  "success": true,
  "message": "Equipment categories retrieved successfully",
  "data": [
    {
      "category_id": 1,
      "category_name": "Audio Visual",
      "description": "Projectors, microphones, speakers"
    },
    {
      "category_id": 2,
      "category_name": "Computing",
      "description": "Laptops, tablets, desktops"
    },
    {
      "category_id": 3,
      "category_name": "Sports Equipment",
      "description": "Balls, nets, athletic gear"
    },
    {
      "category_id": 4,
      "category_name": "Laboratory Equipment",
      "description": "Scientific instruments and tools"
    },
    {

```

```
"category_id": 5,
"category_name": "Photography",
"description": "Cameras, lenses, lighting equipment"
}
]
}
```

RESERVATIONS

GET /api/reservations

Request:

Query Parameters:

page: integer (optional, default: 1)
limit: integer (optional, default: 10)
status_id: integer (optional)
facility_id: integer (optional)
user_id: integer (optional, admin only)
start_date: string (optional, format: YYYY-MM-DD)
end_date: string (optional, format: YYYY-MM-DD)

Headers:

Authorization: Bearer {token}

Response (200 OK) - For Student/Organization:

```
json
{
  "success": true,
  "message": "Reservations retrieved successfully",
  "data": {
    "reservations": [
      {
        "reservation_id": 45,
        "facility_id": 1,
        "facility_name": "Conference Room A",
        "facility_type": "Conference Room",
        "reservation_date": "2025-10-10",
        "start_time": "09:00:00",
        "end_time": "11:00:00",

```

```

    "participant_count": 25,
    "status_id": 2,
    "status_name": "Approved",
    "admin_notes": null,
    "created_at": "2025-10-05T14:30:00Z",
    "reviewed_at": "2025-10-06T09:00:00Z"
  },
  {
    "reservation_id": 48,
    "facility_id": 2,
    "facility_name": "Basketball Court",
    "facility_type": "Sports Facility",
    "reservation_date": "2025-10-12",
    "start_time": "15:00:00",
    "end_time": "17:00:00",
    "participant_count": 15,
    "status_id": 1,
    "status_name": "Pending",
    "admin_notes": null,
    "created_at": "2025-10-08T10:15:00Z",
    "reviewed_at": null
  }
],
"pagination": {
  "current_page": 1,
  "total_pages": 2,
  "total_reservations": 12,
  "per_page": 10
}
}
}

```

Response (200 OK) - For Admin:

```

json
{
  "success": true,
  "message": "Reservations retrieved successfully",
  "data": {
    "reservations": [
      {
        "reservation_id": 45,
        "user_id": 15,
        "user_name": "Jane Smith",

```

```
    "user_email": "jane.smith@example.com",
    "user_role": "Student",
    "facility_id": 1,
    "facility_name": "Conference Room A",
    "facility_type": "Conference Room",
    "reservation_date": "2025-10-10",
    "start_time": "09:00:00",
    "end_time": "11:00:00",
    "participant_count": 25,
    "status_id": 2,
    "status_name": "Approved",
    "admin_notes": "Approved for thesis defense",
    "created_at": "2025-10-05T14:30:00Z",
    "reviewed_at": "2025-10-06T09:00:00Z",
    "updated_at": "2025-10-06T09:00:00Z"
  }
],
"pagination": {
  "current_page": 1,
  "total_pages": 8,
  "total_reservations": 78,
  "per_page": 10
}
}
```

GET /api/reservations/:id

Request:

Path Parameter:

id: integer (required)

Headers:

Authorization: Bearer {token}

Response (200 OK):

json

```
{  
  "success": true,  
  "message": "Reservation retrieved successfully",  
  "data": {  
    "reservation_id": 45,  
    "user_id": 15,  
    "user_name": "Jane Smith",  
    "user_email": "jane.smith@example.com",  
    "user_phone": "+639123456789",  
    "facility_id": 1,  
    "facility_name": "Conference Room A",  
    "facility_type": "Conference Room",  
    "facility_capacity": 50,  
    "reservation_date": "2025-10-10",  
    "start_time": "09:00:00",  
    "end_time": "11:00:00",  
    "participant_count": 25,  
    "status_id": 2,  
    "status_name": "Approved",  
    "admin_notes": "Approved for thesis defense",  
    "created_at": "2025-10-05T14:30:00Z",  
    "reviewed_at": "2025-10-06T09:00:00Z",  
    "updated_at": "2025-10-06T09:00:00Z"  
  }  
}
```

```
}
```

POST /api/reservations

Request:

json

```
{  
  "facility_id": "integer (required)",  
  "reservation_date": "string (required, format: YYYY-MM-DD)",  
  "start_time": "string (required, format: HH:MM:SS)",  
  "end_time": "string (required, format: HH:MM:SS)",  
  "participant_count": "integer (optional)"  
}
```

Response (201 Created):

json

```
{  
  "success": true,  
  "message": "Reservation created successfully",  
  "data": {  
    "reservation_id": 52,  
    "facility_id": 1,  
    "facility_name": "Conference Room A",  
    "reservation_date": "2025-10-15",  
    "start_time": "14:00:00",  
    "end_time": "16:00:00",
```

```
"participant_count": 30,  
"status_id": 1,  
"status_name": "Pending",  
"created_at": "2025-10-09T18:30:00Z"  
}  
}
```

Response (400 Bad Request - Time slot not available):

```
json  
{  
  "success": false,  
  "message": "Time slot is already reserved",  
  "data": {  
    "conflicting_reservation": {  
      "reservation_id": 45,  
      "start_time": "13:00:00",  
      "end_time": "15:00:00"  
    }  
  }  
}
```

Response (400 Bad Request - Past date):

```
json  
{  
  "success": false,  
  "message": "Cannot create reservation for past dates",
```

```
"data": null
}
```

PATCH /api/reservations/:id

Request:

json

```
{
  "reservation_date": "string (optional, format: YYYY-MM-DD)",
  "start_time": "string (optional, format: HH:MM:SS)",
  "end_time": "string (optional, format: HH:MM:SS)",
  "participant_count": "integer (optional)"
}
```

Response (200 OK):

json

```
{
  "success": true,
  "message": "Reservation updated successfully",
  "data": {
    "reservation_id": 52,
    "facility_id": 1,
    "facility_name": "Conference Room A",
    "reservation_date": "2025-10-15",
    "start_time": "15:00:00",
    "end_time": "17:00:00",
  }
}
```



```
"participant_count": 35,  
"status_id": 1,  
"status_name": "Pending",  
"updated_at": "2025-10-09T19:00:00Z"  
}  
}
```

Response (403 Forbidden - Cannot edit approved/declined):

json

```
{  
  "success": false,  
  "message": "Cannot edit reservation with status: Approved",  
  "data": null  
}
```

DELETE /api/reservations/:id

Request:

Path Parameter:

id: integer (required)

Headers:

Authorization: Bearer {token}

Response (200 OK):

json

```
{
```

```
"success": true,  
"message": "Reservation cancelled successfully",  
"data": {  
  "reservation_id": 52,  
  "status_id": 4,  
  "status_name": "Cancelled"  
}  
}
```

Response (403 Forbidden):

```
json  
{  
  "success": false,  
  "message": "Cannot cancel reservation that starts in less than 24 hours",  
  "data": null  
}
```

PATCH /api/reservations/:id/approve (*Admin only*)

Request:

```
json  
{  
  "admin_notes": "string (optional)"  
}
```

Response (200 OK):

```
json
```

```
{
  "success": true,
  "message": "Reservation approved successfully",
  "data": {
    "reservation_id": 48,
    "status_id": 2,
    "status_name": "Approved",
    "admin_notes": "Approved for club meeting",
    "reviewed_at": "2025-10-09T19:30:00Z"
  }
}
```

PATCH /api/reservations/:id/decline (*Admin only*)

Request:

json

```
{
  "admin_notes": "string (required)"
}
```

Response (200 OK):

json

```
{
  "success": true,
  "message": "Reservation declined successfully",
  "data": {
```

```
"reservation_id": 50,  
"status_id": 3,  
"status_name": "Declined",  
"admin_notes": "Facility under maintenance on requested date",  
"reviewed_at": "2025-10-09T19:35:00Z"  
}  
}
```

GET /api/reservations/history

Request:

Query Parameters:

page: integer (optional, default: 1)
limit: integer (optional, default: 10)
start_date: string (optional, format: YYYY-MM-DD)
end_date: string (optional, format: YYYY-MM-DD)
status_id: integer (optional)

Headers:

Authorization: Bearer {token}

Response (200 OK):

json

```
{  
  "success": true,  
  "message": "Reservation history retrieved successfully",
```

```
"data": {  
  "reservations": [  
    {  
      "reservation_id": 23,  
      "facility_id": 1,  
      "facility_name": "Conference Room A",  
      "reservation_date": "2025-09-15",  
      "start_time": "10:00:00",  
      "end_time": "12:00:00",  
      "status_id": 2,  
      "status_name": "Approved",  
      "created_at": "2025-09-10T08:00:00Z",  
      "reviewed_at": "2025-09-11T09:00:00Z"  
    },  
    {  
      "reservation_id": 18,  
      "facility_id": 3,  
      "facility_name": "Study Room C",  
      "reservation_date": "2025-08-20",  
      "start_time": "14:00:00",  
      "end_time": "16:00:00",  
      "status_id": 5,  
      "status_name": "Completed",  
      "created_at": "2025-08-15T10:00:00Z",
```

```
    "reviewed_at": "2025-08-16T11:00:00Z"
  }
],
"pagination": {
  "current_page": 1,
  "total_pages": 3,
  "total_reservations": 28,
  "per_page": 10
},
"statistics": {
  "total_reservations": 28,
  "approved": 20,
  "declined": 5,
  "cancelled": 3
}
}
```

BORROWING

GET /api/borrowing

Request:

Query Parameters:

page: integer (optional, default: 1)

limit: integer (optional, default: 10)

status_id: integer (optional)
equipment_id: integer (optional)
user_id: integer (optional, admin only)
start_date: string (optional, format: YYYY-MM-DD)
end_date: string (optional, format: YYYY-MM-DD)

Headers:

Authorization: Bearer {token}

Response (200 OK) - For Student/Organization:

json

```
{  
  "success": true,  
  "message": "Borrowing requests retrieved successfully",  
  "data": {  
    "borrowings": [  
      {  
        "borrowing_id": 23,  
        "equipment_id": 1,  
        "equipment_name": "Projector",  
        "equipment_code": "PROJ-001",  
        "category_name": "Audio Visual",  
        "borrow_date": "2025-10-08",  
        "expected_return_date": "2025-10-12",  
        "actual_return_date": null,  

```

```
"status_id": 2,  
"status_name": "Approved",  
"condition_notes": "In good condition",  
"admin_notes": null,  
"created_at": "2025-10-05T10:00:00Z",  
"approved_at": "2025-10-06T08:30:00Z"  
},  
{  
  "borrowing_id": 28,  
  "equipment_id": 5,  
  "equipment_name": "Camera",  
  "equipment_code": "CAM-003",  
  "category_name": "Photography",  
  "borrow_date": "2025-10-11",  
  "expected_return_date": "2025-10-15",  
  "actual_return_date": null,  
  "status_id": 1,  
  "status_name": "Pending",  
  "condition_notes": "Need for event coverage",  
  "admin_notes": null,  
  "created_at": "2025-10-09T14:20:00Z",  
  "approved_at": null  
}  
],
```



```
"pagination": {  
  "current_page": 1,  
  "total_pages": 2,  
  "total_borrowings": 15,  
  "per_page": 10  
}  
}  
}
```

Response (200 OK) - For Admin:

json

```
{  
  "success": true,  
  "message": "Borrowing requests retrieved successfully",  
  "data": {  
    "borrowings": [  
      {  
        "borrowing_id": 23,  
        "user_id": 15,  
        "user_name": "Jane Smith",  
        "user_email": "jane.smith@example.com",  
        "user_role": "Student",  
        "equipment_id": 1,  
        "equipment_name": "Projector",  
        "equipment_code": "PROJ-001",
```

```
"category_name": "Audio Visual",
"borrow_date": "2025-10-08",
"expected_return_date": "2025-10-12",
"actual_return_date": null,
"status_id": 2,
"status_name": "Approved",
"approved_by": 1,
"approved_by_name": "Admin User",
"condition_notes": "In good condition",
"admin_notes": "Approved for class presentation",
"created_at": "2025-10-05T10:00:00Z",
"approved_at": "2025-10-06T08:30:00Z",
"updated_at": "2025-10-06T08:30:00Z"
}
],
"pagination": {
  "current_page": 1,
  "total_pages": 12,
  "total_borrowings": 115,
  "per_page": 10
}
}
```

GET /api/borrowing/:id

Request:

Path Parameter:

id: integer (required)

Headers:

Authorization: Bearer {token}

Response (200 OK):

json

```
{  
  "success": true,  
  "message": "Borrowing request retrieved successfully",  
  "data": {  
    "borrowing_id": 23,  
    "user_id": 15,  
    "user_name": "Jane Smith",  
    "user_email": "jane.smith@example.com",  
    "user_phone": "+639123456789",  
    "equipment_id": 1,  
    "equipment_name": "Projector",  
    "equipment_code": "PROJ-001",  
    "category_name": "Audio Visual",  
    "brand_model": "Epson EB-X05",  
    "borrow_date": "2025-10-08",  
  }  
}
```

```
"expected_return_date": "2025-10-12",
"actual_return_date": null,
"status_id": 2,
"status_name": "Approved",
"approved_by": 1,
"approved_by_name": "Admin User",
"condition_notes": "In good condition, all accessories included",
"admin_notes": "Approved for class presentation",
"created_at": "2025-10-05T10:00:00Z",
"approved_at": "2025-10-06T08:30:00Z",
"updated_at": "2025-10-06T08:30:00Z"
}
}
```

POST /api/borrowing

Request:

json

```
{
  "equipment_id": "integer (required)",
  "borrow_date": "string (required, format: YYYY-MM-DD)",
  "expected_return_date": "string (required, format: YYYY-MM-DD)",
  "condition_notes": "string (optional)"
}
```

Response (201 Created):

json

```
{  
  "success": true,  
  "message": "Borrowing request created successfully",  
  "data": {  
    "borrowing_id": 32,  
    "equipment_id": 1,  
    "equipment_name": "Projector",  
    "borrow_date": "2025-10-14",  
    "expected_return_date": "2025-10-18",  
    "status_id": 1,  
    "status_name": "Pending",  
    "condition_notes": "Need for workshop",  
    "created_at": "2025-10-09T20:00:00Z"  
  }  
}
```

Response (400 Bad Request - Equipment not available):

json

```
{  
  "success": false,  
  "message": "Equipment not available for the requested dates",  
  "data": {  
    "available_quantity": 0,  
    "next_available_date": "2025-10-20"  
  }  
}
```

```
}  
}
```

PATCH /api/borrowing/:id

Request:

json

```
{  
  "expected_return_date": "string (optional, format: YYYY-MM-DD)",  
  "condition_notes": "string (optional)"  
}
```

Response (200 OK):

json

```
{  
  "success": true,  
  "message": "Borrowing request updated successfully",  
  "data": {  
    "borrowing_id": 32,  
    "equipment_id": 1,  
    "equipment_name": "Projector",  
    "borrow_date": "2025-10-14",  
    "expected_return_date": "2025-10-19",  
    "status_id": 1,  
    "status_name": "Pending",  
    "condition_notes": "Need for extended workshop",
```

```
"updated_at": "2025-10-09T20:30:00Z"
}
}
```

DELETE /api/borrowing/:id

Request:

Path Parameter:

id: integer (required)

Headers:

Authorization: Bearer {token}

Response (200 OK):

json

```
{
  "success": true,
  "message": "Borrowing request cancelled successfully",
  "data": {
    "borrowing_id": 32,
    "status_id": 5,
    "status_name": "Cancelled"
  }
}
```

PATCH /api/borrowing/:id/approve (*Admin only*)

Request:

json

```
{  
  "admin_notes": "string (optional)"  
}
```

Response (200 OK):

json

```
{  
  "success": true,  
  "message": "Borrowing request approved successfully",  
  "data": {  
    "borrowing_id": 28,  
    "status_id": 2,  
    "status_name": "Approved",  
    "approved_by": 1,  
    "approved_by_name": "Admin User",  
    "admin_notes": "Approved for event coverage",  
    "approved_at": "2025-10-09T20:45:00Z"  
  }  
}
```

PATCH /api/borrowing/:id/decline (*Admin only*)

Request:

json


```
{  
  "admin_notes": "string (required)"  
}
```

Response (200 OK):

json

```
{  
  "success": true,  
  "message": "Borrowing request declined successfully",  
  "data": {  
    "borrowing_id": 30,  
    "status_id": 3,  
    "status_name": "Declined",  
    "admin_notes": "Equipment under maintenance",  
    "approved_at": "2025-10-09T20:50:00Z"  
  }  
}
```

PATCH /api/borrowing/:id/return (*Admin only*)

Request:

json

```
{  
  "actual_return_date": "string (required, format: YYYY-MM-DD)",  
  "condition_notes": "string (optional)"  
}
```

Response (200 OK):

json

```
{
  "success": true,
  "message": "Equipment marked as returned successfully",
  "data": {
    "borrowing_id": 23,
    "equipment_id": 1,
    "equipment_name": "Projector",
    "status_id": 4,
    "status_name": "Returned",
    "borrow_date": "2025-10-08",
    "expected_return_date": "2025-10-12",
    "actual_return_date": "2025-10-11",
    "condition_notes": "Returned in good condition",
    "is_late": false,
    "updated_at": "2025-10-11T16:00:00Z"
  }
}
```

Response (200 OK - Late return):

json

```
{
  "success": true,
  "message": "Equipment marked as returned (OVERDUE)",
```

```
"data": {  
  "borrowing_id": 25,  
  "equipment_id": 2,  
  "equipment_name": "Laptop",  
  "status_id": 4,  
  "status_name": "Returned",  
  "borrow_date": "2025-10-01",  
  "expected_return_date": "2025-10-05",  
  "actual_return_date": "2025-10-09",  
  "condition_notes": "Returned late, minor scratches",  
  "is_late": true,  
  "days_overdue": 4,  
  "updated_at": "2025-10-09T21:00:00Z"  
}  
}
```

GET /api/borrowing/history

Request:

Query Parameters:

page: integer (optional, default: 1)

limit: integer (optional, default: 10)

start_date: string (optional, format: YYYY-MM-DD)

end_date: string (optional, format: YYYY-MM-DD)

status_id: integer (optional)

Headers:

Authorization: Bearer {token}

Response (200 OK):

json

```
{
  "success": true,
  "message": "Borrowing history retrieved successfully",
  "data": {
    "borrowings": [
      {
        "borrowing_id": 15,
        "equipment_id": 1,
        "equipment_name": "Projector",
        "borrow_date": "2025-09-10",
        "expected_return_date": "2025-09-14",
        "actual_return_date": "2025-09-13",
        "status_id": 4,
        "status_name": "Returned",
        "is_late": false,
        "created_at": "2025-09-05T08:00:00Z"
      },
      {
        "borrowing_id": 8,
```

```
"equipment_id": 5,
"equipment_name": "Camera",
"borrow_date": "2025-08-15",
"expected_return_date": "2025-08-18",
"actual_return_date": "2025-08-22",
"status_id": 4,
"status_name": "Returned",
"is_late": true,
"days_overdue": 4,
"created_at": "2025-08-10T10:00:00Z"
}
],
"pagination": {
  "current_page": 1,
  "total_pages": 2,
  "total_borrowings": 18,
  "per_page": 10
},
"statistics": {
  "total_borrowings": 18,
  "approved": 15,
  "declined": 2,
  "cancelled": 1,
  "late_returns": 3
```

```
}  
}  
}
```

NOTIFICATIONS

GET /api/notifications

Request:

Query Parameters:

page: integer (optional, default: 1)

limit: integer (optional, default: 20)

is_read: boolean (optional)

notification_type: string (optional)

Headers:

Authorization: Bearer {token}

Response (200 OK):

json

```
{  
  "success": true,  
  "message": "Notifications retrieved successfully",  
  "data": {  
    "notifications": [  
      {
```

```
"notification_id": 45,

"notification_type": "reservation_approved",

"title": "Reservation Approved",

"message": "Your reservation for Conference Room A on 2025-10-10 has been
approved",

"is_read": false,

"reference_type": "reservation",

"reference_id": 45,

"created_at": "2025-10-06T09:00:00Z",

"read_at": null

},

{

"notification_id": 44,

"notification_type": "borrowing_reminder",

"title": "Return Reminder",

"message": "Your borrowed Projector is due for return tomorrow (2025-10-12)",

"is_read": true,

"reference_type": "borrowing",

"reference_id": 23,

"created_at": "2025-10-11T08:00:00Z",

"read_at": "2025-10-11T10:30:00Z"

},

{

"notification_id": 42,

"notification_type": "reservation_declined",
```

```
    "title": "Reservation Declined",
    "message": "Your reservation request has been declined. Reason: Facility under maintenance",
    "is_read": true,
    "reference_type": "reservation",
    "reference_id": 50,
    "created_at": "2025-10-09T19:35:00Z",
    "read_at": "2025-10-09T20:00:00Z"
  }
],
"pagination": {
  "current_page": 1,
  "total_pages": 3,
  "total_notifications": 52,
  "per_page": 20
},
"unread_count": 8
}
```

PATCH /api/notifications/:id/read

Request:

Path Parameter:

id: integer (required)

Headers:

Authorization: Bearer {token}

Response (200 OK):

json

```
{  
  "success": true,  
  "message": "Notification marked as read",  
  "data": {  
    "notification_id": 45,  
    "is_read": true,  
    "read_at": "2025-10-09T21:30:00Z"  
  }  
}
```

PATCH /api/notifications/read-all

Request:

Headers:

Authorization: Bearer {token}

Response (200 OK):

json

```
{  
  "success": true,  
  "message": "All notifications marked as read",  
  "data": {
```

```
"marked_count": 8,  
"read_at": "2025-10-09T21:35:00Z"  
}  
}
```

REPORTS & LOGS

GET /api/reports/dashboard (*Admin only*)

Request:

Query Parameters:

date_range: string (optional, values: "today", "week", "month", "year")

Headers:

Authorization: Bearer {admin_token}

Response (200 OK):

json

```
{  
  "success": true,  
  "message": "Dashboard statistics retrieved successfully",  
  "data": {  
    "overview": {  
      "total_users": 245,  
      "active_users": 198,  
      "total_facilities": 25,  
      "available_facilities": 22,
```

```
"total_equipment": 150,
"available_equipment": 132
},
"reservations": {
  "total": 450,
  "pending": 15,
  "approved": 320,
  "declined": 45,
  "cancelled": 35,
  "completed": 335,
  "upcoming": 18
},
"borrowings": {
  "total": 680,
  "pending": 12,
  "approved": 25,
  "active": 38,
  "returned": 580,
  "overdue": 8,
  "declined": 17
},
"recent_activity": [
  {
    "type": "reservation",
```

```
"action": "created",
"user_name": "Jane Smith",
"description": "Reserved Conference Room A",
"timestamp": "2025-10-09T20:00:00Z"
},
{
  "type": "borrowing",
  "action": "approved",
  "admin_name": "Admin User",
  "description": "Approved Camera borrowing for John Doe",
  "timestamp": "2025-10-09T19:45:00Z"
}
],
"popular_facilities": [
  {
    "facility_id": 1,
    "facility_name": "Conference Room A",
    "reservation_count": 85
  },
  {
    "facility_id": 5,
    "facility_name": "Computer Lab 1",
    "reservation_count": 72
  }
]
```

```
],  
  "popular_equipment": [  
    {  
      "equipment_id": 1,  
      "equipment_name": "Projector",  
      "borrow_count": 142  
    },  
    {  
      "equipment_id": 2,  
      "equipment_name": "Laptop",  
      "borrow_count": 95  
    }  
  ]  
}  
}
```

GET /api/reports/audit-logs (*Admin only*)

Request:

Query Parameters:

page: integer (optional, default: 1)
limit: integer (optional, default: 50)
start_date: string (optional, format: YYYY-MM-DD)
end_date: string (optional, format: YYYY-MM-DD)
user_id: integer (optional)

action: string (optional)

table_name: string (optional)

Headers:

Authorization: Bearer {admin_token}

Response (200 OK):

json

```
{
  "success": true,
  "message": "Audit logs retrieved successfully",
  "data": {
    "logs": [
      {
        "log_id": 1523,
        "user_id": 15,
        "user_email": "jane.smith@example.com",
        "action": "CREATE",
        "table_name": "FACILITY_RESERVATION",
        "record_id": 52,
        "old_values": null,
        "new_values": "{\"facility_id\": 1, \"reservation_date\": \"2025-10-15\", \"start_time\": \"14:00:00\"}",
        "ip_address": "192.168.1.100",
        "created_at": "2025-10-09T18:30:00Z"
      }
    ]
  }
}
```

```
{
  "log_id": 1522,
  "user_id": 1,
  "user_email": "admin@example.com",
  "action": "UPDATE",
  "table_name": "FACILITY_RESERVATION",
  "record_id": 48,
  "old_values": "{\"status_id\": 1}",
  "new_values": "{\"status_id\": 2, \"admin_notes\": \"Approved\"}",
  "ip_address": "192.168.1.50",
  "created_at": "2025-10-09T19:30:00Z"
},
{
  "log_id": 1521,
  "user_id": 16,
  "user_email": "org@example.com",
  "action": "CREATE",
  "table_name": "EQUIPMENT_BORROWING",
  "record_id": 28,
  "old_values": null,
  "new_values": "{\"equipment_id\": 5, \"borrow_date\": \"2025-10-11\"}",
  "ip_address": "192.168.1.105",
  "created_at": "2025-10-09T14:20:00Z"
}
```

```
],  
  "pagination": {  
    "current_page": 1,  
    "total_pages": 31,  
    "total_logs": 1523,  
    "per_page": 50  
  }  
}  
}
```

GET /api/reports/overdue (*Admin only*)

Request:

Query Parameters:

page: integer (optional, default: 1)

limit: integer (optional, default: 20)

Headers:

Authorization: Bearer {admin_token}

Response (200 OK):

json

```
{  
  "success": true,  
  "message": "Overdue borrowings retrieved successfully",  
  "data": {
```



```
"overdue_borrowings": [  
  {  
    "borrowing_id": 18,  
    "user_id": 22,  
    "user_name": "Mike Johnson",  
    "user_email": "mike.j@example.com",  
    "user_phone": "+639123456780",  
    "equipment_id": 3,  
    "equipment_name": "DSLR Camera",  
    "equipment_code": "CAM-001",  
    "borrow_date": "2025-09-28",  
    "expected_return_date": "2025-10-05",  
    "days_overdue": 4,  
    "status_id": 2,  
    "status_name": "Approved"  
  },  
  {  
    "borrowing_id": 20,  
    "user_id": 35,  
    "user_name": "Sarah Lee",  
    "user_email": "sarah.l@example.com",  
    "user_phone": "+639123456781",  
    "equipment
```

Retry

Claude does not have the ability to run the code it generates yet.

K

Continue

json

```
"equipment_id": 7,
"equipment_name": "Tablet",
"equipment_code": "TAB-002",
"borrow_date": "2025-10-01",
"expected_return_date": "2025-10-07",
"days_overdue": 2,
"status_id": 2,
"status_name": "Approved"
}
],
"pagination": {
  "current_page": 1,
  "total_pages": 1,
  "total_overdue": 8,
  "per_page": 20
},
"summary": {
  "total_overdue": 8,
  "total_overdue_value": 12500.00,
  "average_days_overdue": 3.5
```

```
}  
}  
}
```

LOOKUPS

GET /api/lookups/roles

Request:

No parameters required

Response (200 OK):

json

```
{  
  "success": true,  
  "message": "Roles retrieved successfully",  
  "data": [  
    {  
      "role_id": 1,  
      "role_name": "Admin"  
    },  
    {  
      "role_id": 2,  
      "role_name": "Organization"  
    },  
    {
```

```
"role_id": 3,  
  "role_name": "Student"  
}  
]  
}
```

GET /api/lookups/user-statuses

Request:

No parameters required

Response (200 OK):

json

```
{  
  "success": true,  
  "message": "User statuses retrieved successfully",  
  "data": [  
    {  
      "status_id": 1,  
      "status_name": "Active"  
    },  
    {  
      "status_id": 2,  
      "status_name": "Inactive"  
    },  
    {
```

```
"status_id": 3,  
  "status_name": "Suspended"  
}  
]  
}
```

GET /api/lookups/reservation-statuses

Request:

No parameters required

Response (200 OK):

json

```
{  
  "success": true,  
  "message": "Reservation statuses retrieved successfully",  
  "data": [  
    {  
      "status_id": 1,  
      "status_name": "Pending"  
    },  
    {  
      "status_id": 2,  
      "status_name": "Approved"  
    },  
    {
```

```
"status_id": 3,
"status_name": "Declined"
},
{
"status_id": 4,
"status_name": "Cancelled"
},
{
"status_id": 5,
"status_name": "Completed"
}
]
}
```

GET /api/lookups/borrowing-statuses

Request:

No parameters required

Response (200 OK):

json

```
{
"success": true,
"message": "Borrowing statuses retrieved successfully",
"data": [
{
```

```
    "status_id": 1,  
    "status_name": "Pending"  
  },  
  {  
    "status_id": 2,  
    "status_name": "Approved"  
  },  
  {  
    "status_id": 3,  
    "status_name": "Declined"  
  },  
  {  
    "status_id": 4,  
    "status_name": "Returned"  
  },  
  {  
    "status_id": 5,  
    "status_name": "Cancelled"  
  },  
  {  
    "status_id": 6,  
    "status_name": "Overdue"  
  }  
]
```

```
}
```

GET /api/lookups/facility-statuses

Request:

No parameters required

Response (200 OK):

json

```
{
  "success": true,
  "message": "Facility statuses retrieved successfully",
  "data": [
    {
      "status_id": 1,
      "status_name": "Available"
    },
    {
      "status_id": 2,
      "status_name": "Under Maintenance"
    },
    {
      "status_id": 3,
      "status_name": "Unavailable"
    }
  ]
}
```



```
}
```

GET /api/lookups/equipment-statuses

Request:

No parameters required

Response (200 OK):

json

```
{
  "success": true,
  "message": "Equipment statuses retrieved successfully",
  "data": [
    {
      "status_id": 1,
      "status_name": "Available"
    },
    {
      "status_id": 2,
      "status_name": "Under Maintenance"
    },
    {
      "status_id": 3,
      "status_name": "Unavailable"
    },
    {
```

```
"status_id": 4,  
"status_name": "Decommissioned"  
}  
]  
}
```

ADDITIONAL HELPFUL ENDPOINTS

GET /api/calendar/reservations (*Calendar View*)

Request:

Query Parameters:

start_date: string (required, format: YYYY-MM-DD)

end_date: string (required, format: YYYY-MM-DD)

facility_id: integer (optional)

Headers:

Authorization: Bearer {token}

Response (200 OK):

json

```
{  
  "success": true,  
  "message": "Calendar reservations retrieved successfully",  
  "data": {  
    "events": [  

```

```
{
  "id": 45,
  "title": "Conference Room A - Jane Smith",
  "facility_id": 1,
  "facility_name": "Conference Room A",
  "start": "2025-10-10T09:00:00Z",
  "end": "2025-10-10T11:00:00Z",
  "status": "Approved",
  "user_name": "Jane Smith",
  "participant_count": 25
},
{
  "id": 48,
  "title": "Basketball Court - Tech Club",
  "facility_id": 2,
  "facility_name": "Basketball Court",
  "start": "2025-10-12T15:00:00Z",
  "end": "2025-10-12T17:00:00Z",
  "status": "Pending",
  "user_name": "Tech Club",
  "participant_count": 15
}
]
```

```
}
```

POST /api/reports/export (*Admin only*)

Request:

json

```
{  
  "report_type": "string (required, values: 'reservations', 'borrowings', 'audit_logs', 'overdue')",  
  "format": "string (required, values: 'csv', 'pdf')",  
  "start_date": "string (optional, format: YYYY-MM-DD)",  
  "end_date": "string (optional, format: YYYY-MM-DD)",  
  "filters": {  
    "status_id": "integer (optional)",  
    "facility_id": "integer (optional)",  
    "equipment_id": "integer (optional)",  
    "user_id": "integer (optional)"  
  }  
}
```

Response (200 OK):

json

```
{  
  "success": true,  
  "message": "Report generated successfully",  
  "data": {  
    "file_url": "https://yourdomain.com/exports/reservations_2025-10-09.csv",
```

```
"file_name": "reservations_2025-10-09.csv",  
"file_size": "245KB",  
"expires_at": "2025-10-10T21:45:00Z"  
}  
}
```

GET /api/users/activity/:id (Admin only)

Request:

Path Parameter:

id: integer (required)

Query Parameters:

page: integer (optional, default: 1)

limit: integer (optional, default: 20)

activity_type: string (optional, values: 'reservation', 'borrowing', 'all')

Headers:

Authorization: Bearer {admin_token}

Response (200 OK):

json

```
{  
  "success": true,  
  "message": "User activity retrieved successfully",  
  "data": {
```

```
"user": {  
  "user_id": 15,  
  "user_name": "Jane Smith",  
  "user_email": "jane.smith@example.com",  
  "role_name": "Student"  
},  
"activity": [  
  {  
    "activity_id": 523,  
    "activity_type": "reservation",  
    "action": "created",  
    "description": "Created reservation for Conference Room A",  
    "reference_id": 52,  
    "timestamp": "2025-10-09T18:30:00Z"  
  },  
  {  
    "activity_id": 485,  
    "activity_type": "borrowing",  
    "action": "returned",  
    "description": "Returned Projector",  
    "reference_id": 23,  
    "timestamp": "2025-10-11T16:00:00Z"  
  }  
],
```

```
"pagination": {
  "current_page": 1,
  "total_pages": 3,
  "total_activities": 58,
  "per_page": 20
},
"statistics": {
  "total_reservations": 32,
  "total_borrowings": 26,
  "pending_requests": 3,
  "completed_requests": 55
}
}
```

POST /api/reservations/:id/extend

Request:

json

```
{
  "new_end_time": "string (required, format: HH:MM:SS)",
  "reason": "string (optional)"
}
```

Response (200 OK):

json

```
{
  "success": true,
  "message": "Extension request submitted successfully",
  "data": {
    "reservation_id": 45,
    "original_end_time": "11:00:00",
    "new_end_time": "13:00:00",
    "extension_status": "pending_approval",
    "reason": "Meeting running longer than expected"
  }
}
```

POST /api/borrowing/:id/extend

Request:

json

```
{
  "new_return_date": "string (required, format: YYYY-MM-DD)",
  "reason": "string (optional)"
}
```

Response (200 OK):

json

```
{
  "success": true,
  "message": "Extension request submitted successfully",
}
```



```
"data": {  
  "borrowing_id": 23,  
  "original_return_date": "2025-10-12",  
  "new_return_date": "2025-10-15",  
  "extension_status": "pending_approval",  
  "reason": "Project deadline extended"  
}  
}
```

GET /api/search

Request:

Query Parameters:

q: string (required, search query)

type: string (optional, values: 'facilities', 'equipment', 'all', default: 'all')

limit: integer (optional, default: 10)

Headers:

Authorization: Bearer {token}

Response (200 OK):

json

```
{  
  "success": true,  
  "message": "Search results retrieved successfully",  
  "data": {  
    "facilities": [  

```

```
{
  "facility_id": 1,
  "facility_name": "Conference Room A",
  "facility_type": "Conference Room",
  "capacity": 50,
  "status": "Available"
},
{
  "equipment_id": 1,
  "equipment_name": "Projector",
  "equipment_code": "PROJ-001",
  "category": "Audio Visual",
  "available_quantity": 7,
  "status": "Available"
},
{
  "total_results": 2
}
```

ERROR RESPONSE FORMATS

All error responses follow a consistent format:

400 Bad Request

json

```
{
  "success": false,
  "message": "Validation error",
  "errors": [
    {
      "field": "email",
      "message": "Email is required"
    },
    {
      "field": "password",
      "message": "Password must be at least 8 characters"
    }
  ]
}
```

401 Unauthorized

json

```
{
  "success": false,
  "message": "Unauthorized access",
  "data": null
}
```

403 Forbidden

json

```
{  
  "success": false,  
  "message": "You do not have permission to perform this action",  
  "data": null  
}
```

404 Not Found

json

```
{  
  "success": false,  
  "message": "Resource not found",  
  "data": null  
}
```

409 Conflict

json

```
{  
  "success": false,  
  "message": "Resource conflict",  
  "data": {  
    "conflict_type": "time_slot_occupied",  
    "details": "Time slot is already reserved"  
  }  
}
```

422 Unprocessable Entity

json

```
{
  "success": false,
  "message": "Cannot process request",
  "data": {
    "reason": "Cannot delete facility with active reservations",
    "active_count": 5
  }
}
```

500 Internal Server Error

json

```
{
  "success": false,
  "message": "An unexpected error occurred",
  "data": null
}
```

HTTP STATUS CODES SUMMARY

Status Code	Description
200 OK	Successful GET, PUT, PATCH, DELETE

201 Created	Successful POST (resource created)
400 Bad Request	Validation errors, invalid input
401 Unauthorized	Missing or invalid token
403 Forbidden	Insufficient permissions
404 Not Found	Resource doesn't exist
409 Conflict	Duplicate resource or conflict
422 Unprocessable Entity	Business logic error
500 Internal Server Error	Server-side error

COMMON HEADERS

Request Headers:

Authorization: Bearer {jwt_token}

Content-Type: application/json

Accept: application/json

Response Headers:

Content-Type: application/json

X-RateLimit-Limit: 100

X-RateLimit-Remaining: 95

X-RateLimit-Reset: 1696867200